

NIRNAY — National Governance & Monitoring Engine

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06

Ministry of Social Justice & Empowerment, Government of India

🚨 Problem Statement (SIH 2025 Challenge)

- **Monitoring Deficit:** Over 10,000+ grantee NGOs, rehabilitation centers, and welfare facilities operating across 700+ districts without real-time tracking.
- **Biometric Fraud:** Static attendance registers lead to proxy attendance, ghost beneficiaries, and fund diversion.
- **Unresolved District Alerts:** SLA delay in field inspection reviews causes administrative bottlenecks.
- **Legal Evidence Gaps:** Inspection evidence lacks admissibility certificates under Indian Evidence Law.



National Decision Support System

Designed by NIC & Gol Hackathon
Squad

💡 Solution: NIRNAY Platform

A Next-Gen e-Governance system integrating **2D/3D Hybrid GIS, Biometric FRS + Live Geofence Lock, AI Anomaly Signals**, and **3-Tier Automated SLA Escalation Engine**.

Core Technical Innovations & Portal Roles

Multi-tier Role-Based Access Control (RBAC) & Attribute Access (ABAC)

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Portal Category	Designated Official	Key Functionality	Security Level
Central Command	Directorate Official	Nationwide 2D/3D GIS surveillance, AI alerts & SLA monitoring	LEVEL 3 (NATIONAL)
Audit Squad	Inspection Officer	Unannounced 10-point audit missions, 2D map GPS geofence lock	FRS + GPS LOCK
Agency Portal	NGO / Institution	Submit daily beneficiary attendance & respond to verification notices	AGENCY AUTH
State District Authority	District Magistrate / IAS	District jurisdiction overview, magistrate report sign-off & issues	REGIONAL MAGISTRATE
System Admin	Platform Administrator	User directory, permission matrix & immutable FRS audit trail	SUPER ADMIN

 Hybrid 2D/3D GIS

2D map for high-performance operational audits; 3D globe mode for command-center analysis.

 FRS & Reverse Geocoding

HTML5 camera facial feature scan paired with live GPS reverse-geocoded address string.

 AI Anomaly Detection

Monitors CCTV stream blackout, ghost attendance, and geofence offset with 94.8% accuracy.

System Architecture & Technical Approach Diagram

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End-to-End System Workflow (SIH 2025 Technical Approach Standard)

 NIRNAY Technical Approach SIH 2025 Workflow Diagram

End-to-End Workflow Execution: Inspector logs in via FRS biometric verification $\rightarrow$ GPS reverse-geocodes exact station address $\rightarrow$ Inspection checklist items hashed with SHA-256 $\rightarrow$ AI Anomaly Engine evaluates risk score $\rightarrow$ SLA Engine auto-escalates unhandled issues to State & Central Admins.

Hierarchical Governance & Automated SLA Escalation

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3-Tier Escalation: District (L1) → State (L2) → Central Command (L3)

Governance Hierarchy

Role + Jurisdiction Scoped Access (RBAC + ABAC):

- **Central Admin:** Nationwide visibility, SLA policy configuration & audit logs.
- **State Admin:** State-scoped (e.g. *Andhra Pradesh*), district escalation reviews.
- **District Admin:** District-scoped (e.g. *Guntur*), inspector team assignment & issues.

SLA Duration Policies

- **CRITICAL (4H SLA)** CCTV Blackout / DVR Tampering
- **HIGH (24H SLA)** Geofence Offset / Biometric Anomaly
- **MEDIUM (72H SLA)** Health Score Drop Below 60%
- **LOW (7D SLA)** Routine Audit Follow-Up

Automated SLA Breach Escalation Flow

When an issue is logged at District level (L1), the SLA clock starts. If unhandled within the specified SLA duration, the **Automated Escalation Engine** automatically promotes the issue to State Admin (L2) and subsequently to Central Command (L3) with internal audit thread tracking.

Legal Evidence Admissibility (BSA 2023 §63B)

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Bharatiya Sakshya Adhiniyam 2023 & DPDP Act 2023 Compliance

Section 63B Admissibility Certificate

Under Bharatiya Sakshya Adhiniyam (BSA) 2023, electronic evidence submitted to courts or administrative tribunals requires a signed **Section 63B Certificate**.

- **SHA-256 Hash Lock:** Cryptographic hash computed at capture time prevents post-inspection tampering.
- **IST & GPS Telemetry:** Indian Standard Time (+05:30) and hardware reverse-geocoded location (*Brodipet, Guntur, AP ±4m*).
- **NIC Device Signature:** Sealed with authorized officer digital signature.

SAMPLE BSA 2023 §63B CERTIFICATE

GOVERNMENT OF INDIA • MINISTRY OF SOCIAL JUSTICE
Admissibility Certificate under BSA 2023 §63B

Evidence ID: EVD-2026-AP-0891
Device ID: NIC-TAB-AP-4402
SHA-256: 8f4b52c79a9e3d81b24e6501a2d718b5...
Location: Brodipet, Guntur, Andhra Pradesh (±4.2m)
Timestamp: 28-Aug-2026 16:10:00 IST

I hereby certify that the electronic record was captured on a lawful device without alteration.
Officer: Priya Mehta (PMU Inspection Squad)

Impact, Scale & Pitch Defense Q&A

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Why NIRNAY Wins Smart India Hackathon (SIH 2025)

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Vitest Test Coverage

63 Routes

Zero Build Errors

100%

5 Government Portals

Key Winning Pitch Differentiators for SIH Jury

- **Complete End-to-End Delivery:** Not just a UI prototype — fully wired state management, issue store, geofencing, and FRS facial camera scanner.
- **Indian E-Governance Standard:** Tailored for Ministry of Social Justice & Empowerment with UX4G design principles and Devanagari branding.
- **Proven Stability:** 100% clean production build and 156 unit test suite.